

Abstract

security implications of deep learning (DL), a machine learning technique for healthcare data analysis. This paper focuses on adversarial attacks, which are digital actions that aim to compromise the accuracy of models, alter the model's output, or worse, reveal sensitive information from the trained models. To defend against adversaries to cause potential damage, this paper explores potential input-based adversarial countermeasures and their associated threats. Furthermore, our study uses selected healthcare datasets to evaluate the effectiveness and interrelatedness of the proposed countermeasures in healthcare data analysis. The results more provide an initial insight into the risks.

Introduction

ning is a machine learning machine learning models by ut. It includes both the of adversarial examples, ially created to deceive Federated Learning) model used for data security in the isbehaves when it faces ssue is seen in the picture



classifies panda images as
prediction of the animal
This illustrates the danger
s represent since we might
e difference. As a result, we
e model's output is valid or

Literature Review

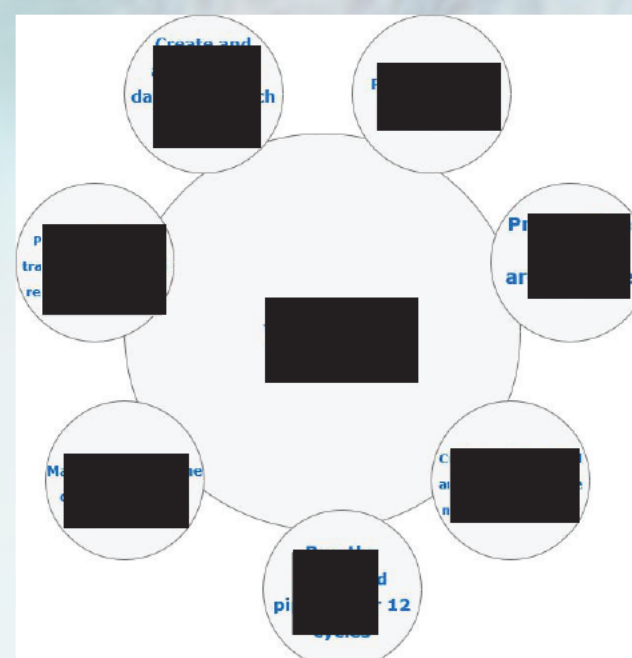
om the paper [2] where generative adversarial g attack. The authors NIST and AT&T) and classify. The result is ing task accuracy can average.

at motivated us to do . Here, the authors attack. They also used 0 and Fashion-MNIST ges. This attack works in which maliciously model by updating the beled data from those

Methodology

The thesis report describes the architecture of the InceptionV3, Federated Learning, and Tools for the analysis of retinal OCT scans. The report also evaluates the model's performance using a confusion matrix, precision-recall curve, and ROC curve. The research aimed to demonstrate the effectiveness of Federated Learning in

Work flow



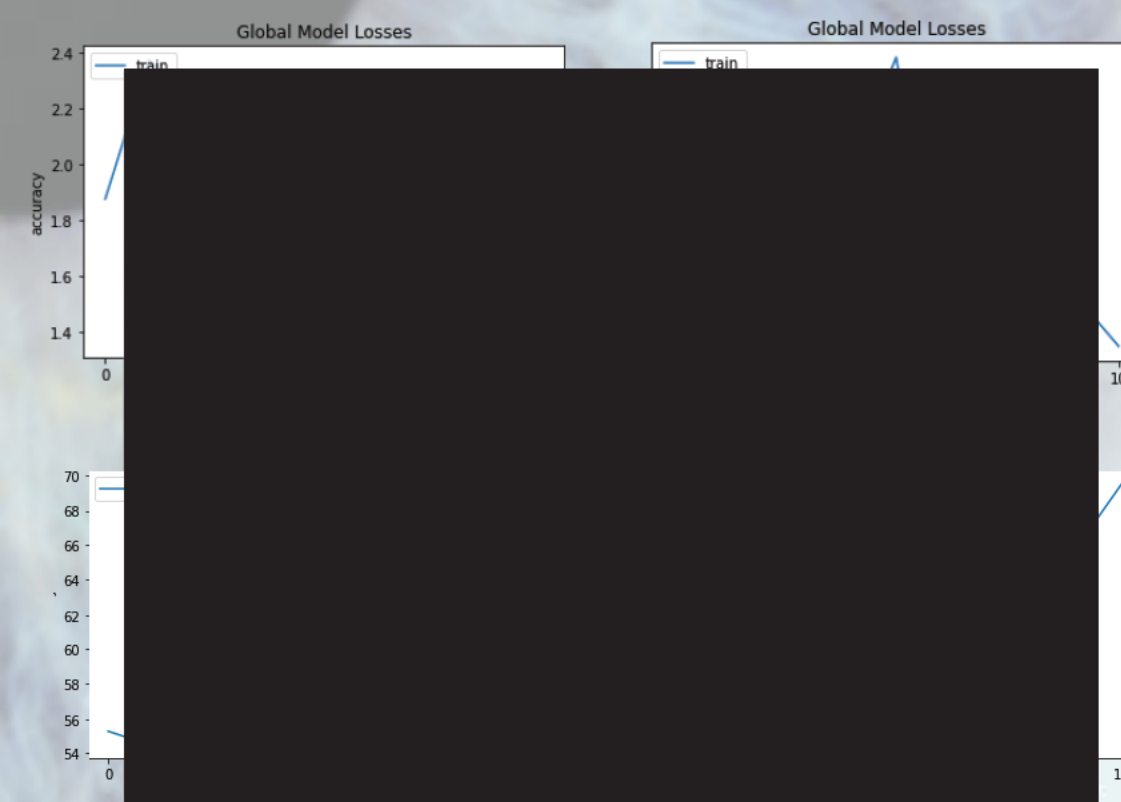
Implementation

valuation of Federated
nd poisoned data is part
Cardiff University. The
pact of regularization
ormance, as well as the
it perform better.

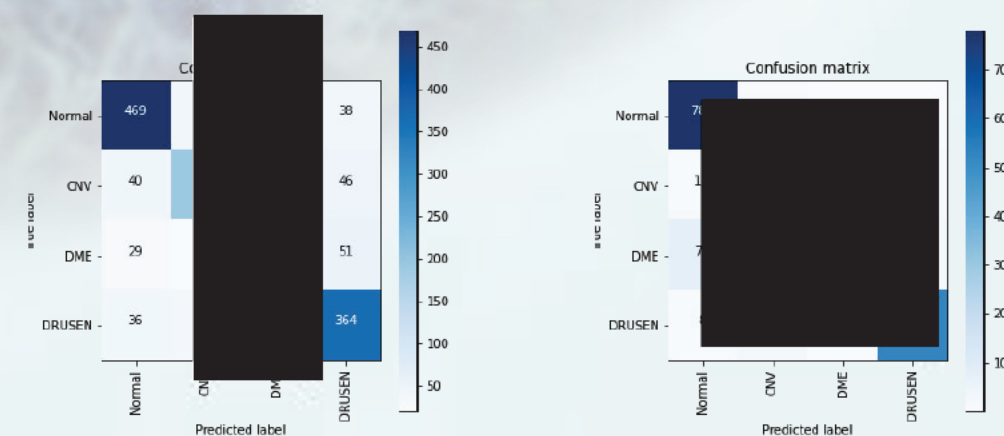
Results

iments on the OCT dataset
ness of data poisoning
Federated Learning. The
e accuracy and loss of the
antly when maliciously
resent in the dataset. A
erated to further support
oncluded deep federated
erable to data poisoning
he use of defense systems

Poisoned



Unpoisoned



Future Work

Our model has shown given dataset. However becomes a problem to processing the computing adversarial attack on the world, then its effect understood. Data poisoning data into the dataset. By attacks like Evasion Attacks. We should use attack too for computing model dataset of (OTC).

Conclusion

In this study, we perform an experiment on the tomography (OCT) data to learn something that it should be able to introduce bad. We established a CNN that successfully performed identifying the images of

References

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